INTERNATIONAL CONFERENCE ON CYBERSECURITY LAW & GOVERNANCE · APRIL 2026

Digital Sovereignty: Lessons from the EU DMA, GDPR, DPDP, and the Path for India

James Reilly · Open Source Advocate & IT Consultant
FOSS United · AlmaLinux · formerly Beeper / Element

The utility question

When did messaging become infrastructure?

- **1850: UK Telegraph Act**

  The first communication utility law. Non-discrimination in sending.

- **2024: EU Digital Markets Act**

  WhatsApp forced open. Same utility principle, 174 years later.

- **India's Reality: 1.4B Users**

  Financial, social, and political lives run on foreign platforms.

What sovereignty actually means

Not isolation. Strategic autonomy.

WHAT IT IS NOT

- Building a Great Firewall
- Banning foreign platforms
- Technological isolationism
- Nationalist protectionism

WHAT IT IS

- The right to audit infrastructure
- Legal jurisdiction over user data
- Ability to switch providers
- Domestic capacity to build alternatives

How Europe built its regulatory shield

Two instruments. One decade. Global impact.

- **GDPR (2018):** Exported European values globally

  Every company serving EU users must comply — regardless of where they're headquartered. India adopted a similar framework with the DPDP Act 2023.

- **Digital Markets Act (2022):** Targeted 'Gatekeepers'

  Apple, Google, Meta, Amazon, Microsoft, ByteDance — forced to open APIs and enable interoperability. Fines up to 10% of global turnover.

- **DMA interoperability in practice:** WhatsApp forced open

  In March 2024, WhatsApp was legally required to allow third-party apps to connect. My former employer Element successfully bridged Matrix to WhatsApp with full E2EE.

Two incidents that changed the conversation

1. The Taurus Leak (Germany)

INCIDENT A · GERMANY 2024

The Taurus Leak (Webex)

Military officers intercepted on an insecure consumer line. Operational failure, not cryptographic.

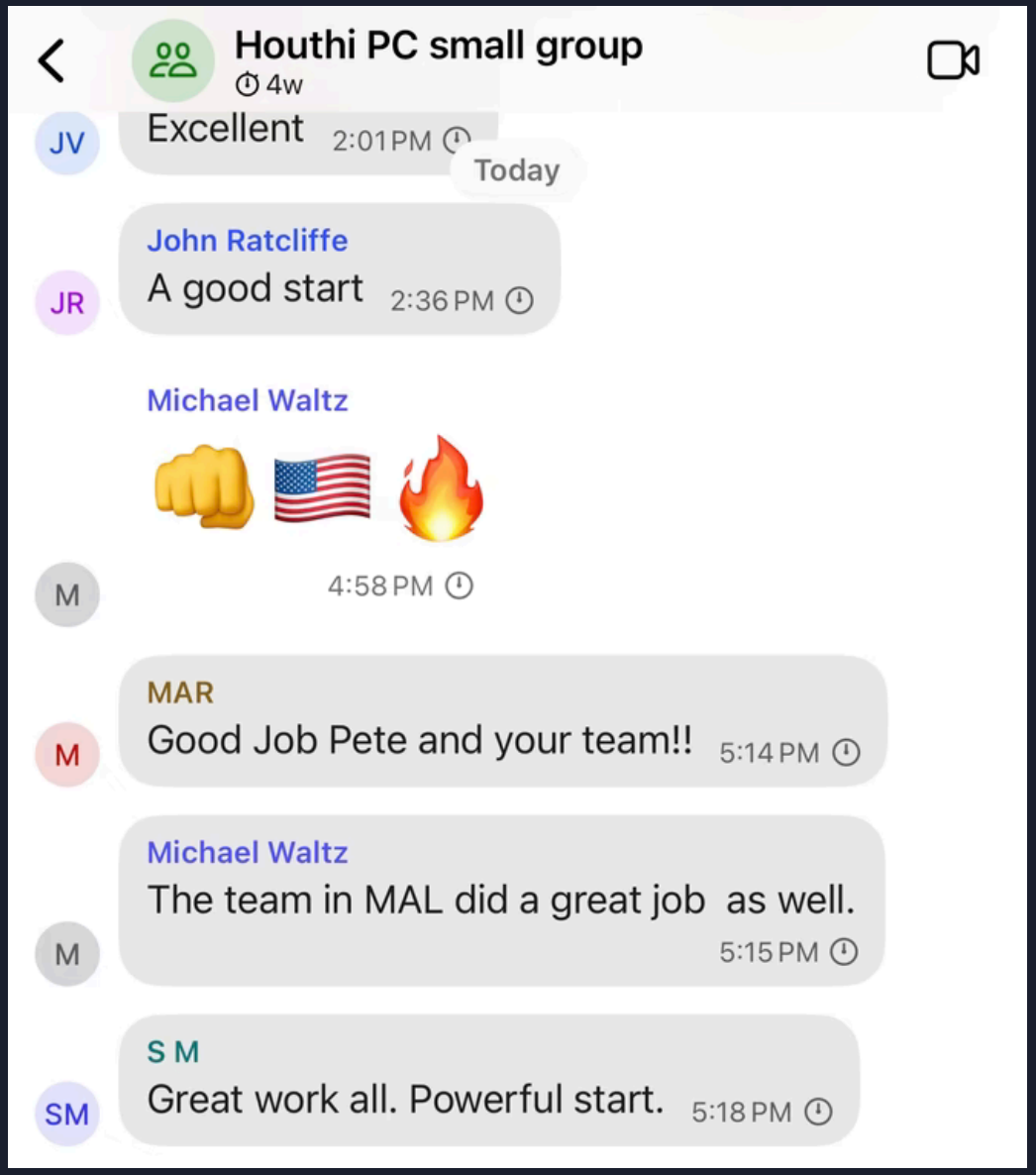
Two incidents that changed the conversation

2. Signalgate (USA)

INCIDENT B · USA 2025

Signalgate (Hegseth / Signal)

Sensitive plans shared via consumer app with a journalist. No audit trail, no access controls.



The sovereignty stack

Law without infrastructure is a promise you can't keep.

04	**Law & Regulation**	DPDP Act, DMA, GDPR — India has this
03	**Data localisation**	Where data physically lives
02	**Infrastructure**	Who runs the servers, clouds, networks
01	**Open Protocols**	Matrix, UPI, SMTP, ActivityPub — the foundation

← India has Layer 4. UPI proved Layer 1 works in payments. The gap is messaging and social media.

India already invented this playbook

UPI, Aadhaar, ONDC — India's open protocol advantage.

- **UPI:** the world's most successful open payments protocol

  20.01 billion transactions in a single month (August 2025). Built on open standards — any bank, any app, true interoperability. Now expanding internationally.

- **ONDC:** open e-commerce, not a marketplace

  A set of open standards for commerce — the same logic as UPI, applied to retail. Any seller, any buyer app, no gatekeeper.

- **The EU envies this. Now apply it to communications.**

  India pioneered open protocols for payments and identity. The gap is messaging and social media — exactly what the DMA is forcing in Europe.

The protocol map: India's wins and gaps

Where open standards exist — and where mandate is still needed.

DOMAIN	OPEN PROTOCOL	STATUS	INDIA POSITION
Messaging	**Matrix / XMPP** Federated, E2EE, open standard	Mandate needed	**No sovereign stack** WhatsApp dominates entirely
Social media	**ActivityPub / AT Protocol** Fediverse, Mastodon, Bluesky, Threads	Mandate needed	**No open alternative** EU expanding DMA here in 2026
Payments	**UPI** Open, interoperable, sovereign	India leads globally	**20.01B txns/month** World's best open payments protocol
E-commerce	**ONDC** Open Network for Digital Commerce	Exists, adoption patchy	**Live since 2022** Same logic as UPI, applied to retail
Identity	**W3C DID / Verifiable Creds** Decentralised identifiers standard	Exists, adoption patchy	**DigiLocker partial** Portable credential standard needed
Mobility	**Beckn Protocol** Open discovery & transaction engine	India leads globally	**Namma Yatri / ONDC** Beating Uber/Ola via open standards

Credit	**OCEN** Open Credit Enablement Network	Exists, adoption patchy	**Live for MSMEs** Breaking the lending gatekeepers
Healthcare	**HL7 FHIR** Fast Healthcare Interoperability Resources	Exists, adoption patchy	**ABDM / UHI** Portable health records & services
Information	**HTTP / HTML (Open Web)** The original level playing field	India leads globally	**Losing to silos** 94% time in apps, 6% in browser
Email	**SMTP / IMAP** The original open protocol	India leads globally	**The baseline example** Gmail ↔ Outlook works. WhatsApp ↔ Signal doesn't — yet.

India proved open protocols beat proprietary gatekeepers in payments. The same playbook — mandate interoperability, fund open infrastructure — applies directly to messaging and social media.

India’s DPDP Act vs GDPR

Strong law — but enforcement infrastructure is missing.

GDPR (EU, 2018)

- Right to access, erasure, portability
- Data Protection Officer mandatory
- Fines up to 4% global revenue
- Extraterritorial reach
- Independent supervisory authorities

DPDP ACT (INDIA, 2023)

- Consent-based framework, similar rights
- Data Protection Board (not yet operational)
- Fines up to ₹250 crore
- Applies to Indian data globally
- Government exemptions remain broad

The human capital advantage

India has the talent. The obstacle is cultural, not technical.

- **Remote work shifted the balance**

  Indian developers can now earn global wages from local cities.

- **The Brand Gap**

  Social prestige still lies with foreign tech giants, posing a cultural obstacle.

- **Open source as equaliser**

  Portable skills built via FOSS (Matrix, Linux) apply directly to sovereign infrastructure solving local problems.

Concrete steps, not aspirations

Each one has a working precedent.

- **Constitute the Data Protection Board — now**

  The DPDP Act exists. Enforcement without an operational board is theatre.

- **Mandate Matrix & ActivityPub for large platforms**

  Require interoperability for any SSMI with 10M+ Indian users. Break the network effect lock-in.

- **Sovereign Govt Infrastructure: Tchap/BundesMessenger model**

  Move all public service communications to an auditable, domestic Matrix-based stack.

- **Upgrade DigiLocker to W3C Verifiable Credentials**

  Move from “fetching PDFs” to cryptographically portable, privacy-preserving identity tokens.

- **Sovereign Education: Mandate FOSS in curricula**

  Stop training customers for foreign firms. Kerala’s KITE model saved ₹3,000 Cr via open standards.

Beyond preaching

Sovereignty is built, not declared.

> *"India doesn't need to ask permission to participate in the global digital economy on its own terms. It has already proven it can build the foundations — with UPI, with Aadhaar, with ONDC. The question is whether it finishes the job."*

THANK YOU

James Reilly

reilly.asia · linkedin.com/in/jreilly112

Open Source Advocate & IT Consultant
FOSS United · AlmaLinux
formerly Beeper (now Automattic) · Element

Questions welcome

Key references & sources

Selected legal and technical foundations.

Legal Foundations

- **Electric Telegraph Company Act 1850** (13 & 14 Vict. c. lxxxvi).
- **Digital Markets Act (DMA)**, Regulation (EU) 2022/1925.
- **GDPR**, Regulation (EU) 2016/679.
- **India DPDP Act**, Act No. 40 of 2023.
- **RBI Directive** on Payment Data Storage (2018).

Technical Standards

- **Matrix protocol** (matrix.org) - IETF/W3C candidate.
- **ActivityPub** (W3C Social Web Working Group).
- **SMTP** (RFC 821/5321).
- **UPI API Specifications** (NPCI).

Reports & Incidents

- **German MoD**, Taurus Leak Investigation (March 2024).
- **The Atlantic**, Signalgate Report (March 2025).
- **NPCI**, UPI Monthly Statistics (August 2025).
- **Meta Newsroom**, Threads ActivityPub Interop (2024).

Sovereign Deployments

- **Tchap** (French Government).
- **BundesMessenger** (German Public Administration).
- **BwMessenger** (German Armed Forces).
- **TI-Messenger** (German Healthcare).